Welcome to RetailCo Australia Your First Day as a Junior Data Analyst

🏢 THE SCENARIO RetailCo Australia

- $500M retail chain
- 150 stores across NSW, VIC, QLD
- Sydney HQ, 3,500 employees

🎯 YOUR MISSION Help the CFO analyze Q4 2025 store performance

THE PROBLEM

Manual Excel reports take 3 days

💰 THE COST

$2,400 per report (analyst time)

📈 THE IMPACT

Slow decisions = lost opportunities

✨ YOUR SOLUTION

Automate analysis with Python in 15 minutes

$28.8K

Annual Savings

12 reports/month automated

15min

Analysis Time

Down from 3 days

📈 BUSINESS IMPACT: Faster decisions = competitive advantage. Your new superpower = Career transformation

The Business Problem Why RetailCo Needs Python

> ## 💼 THE CFO'S QUESTION
>
> "Which stores are underperforming and why?"

📊 CURRENT SITUATION

- Data trapped in Excel files
- Manual analysis = 3 days per report
- Inconsistent criteria across reviews
- Delayed intervention on problems

💸 THE COST OF DELAYS

- 3 days × $800/day analyst time = $2,400/report
- Late problem detection = 15% revenue loss
- Manual errors = wrong business decisions

🎯 WHAT PYTHON SOLVES

- Automate repetitive analysis
- Process 150 stores in 2 seconds
- Consistent evaluation criteria
- Immediate insights, not 3-day delays

📈 THE OPPORTUNITY $28,800 annual savings PLUS faster revenue recovery

Why Python? (Not Excel)

The Tool That Changes Everything

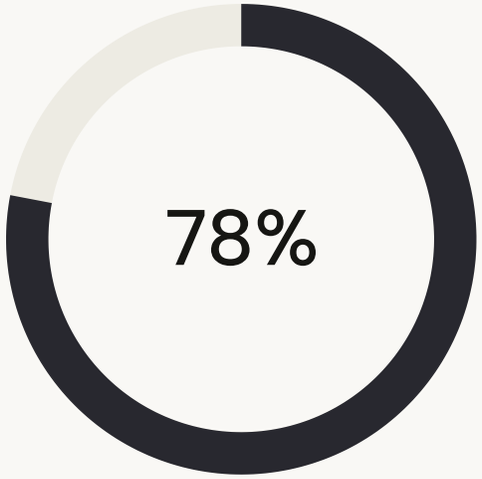
📉 EXCEL LIMITATIONS

- 1,048,576 row maximum
- Crashes with large datasets
- Manual copy-paste prone to errors
- No true automation (macros break)
- Single-user, not collaborative

🐍 PYTHON SUPERPOWERS

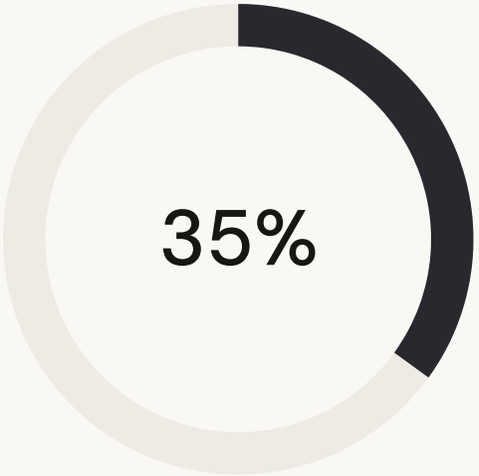
- Unlimited data processing
- True automation (runs without you)
- Scales infinitely (same code, any size)
- Industry-standard enterprise tool
- Free and open-source

💼 INDUSTRY REALITY



Data Jobs Require Python

LinkedIn 2025 data



Higher Earnings

Python vs Excel-only analysts

Top companies: Google, Tesla, Amazon all use Python

Australian market: High demand, low supply

🚀 RETAILCO'S TECH STACK

Python → SQL → Power BI (You're learning the foundation today!)

💰 YOUR CAREER IMPACT

Excel analyst: $75K salary ceiling

Python analyst: $100K+ earning potential

Setting Up Google Colab

Your Free Cloud Python Environment

01

🌐 WHAT IS GOOGLE COLAB?

- Free cloud-based Python environment
- No installation required
- Runs in your web browser
- Powered by Google's infrastructure

02

🔗 GET STARTED

Visit: colab.research.google.com

Sign in with Google account

Click: New Notebook

Start coding immediately!

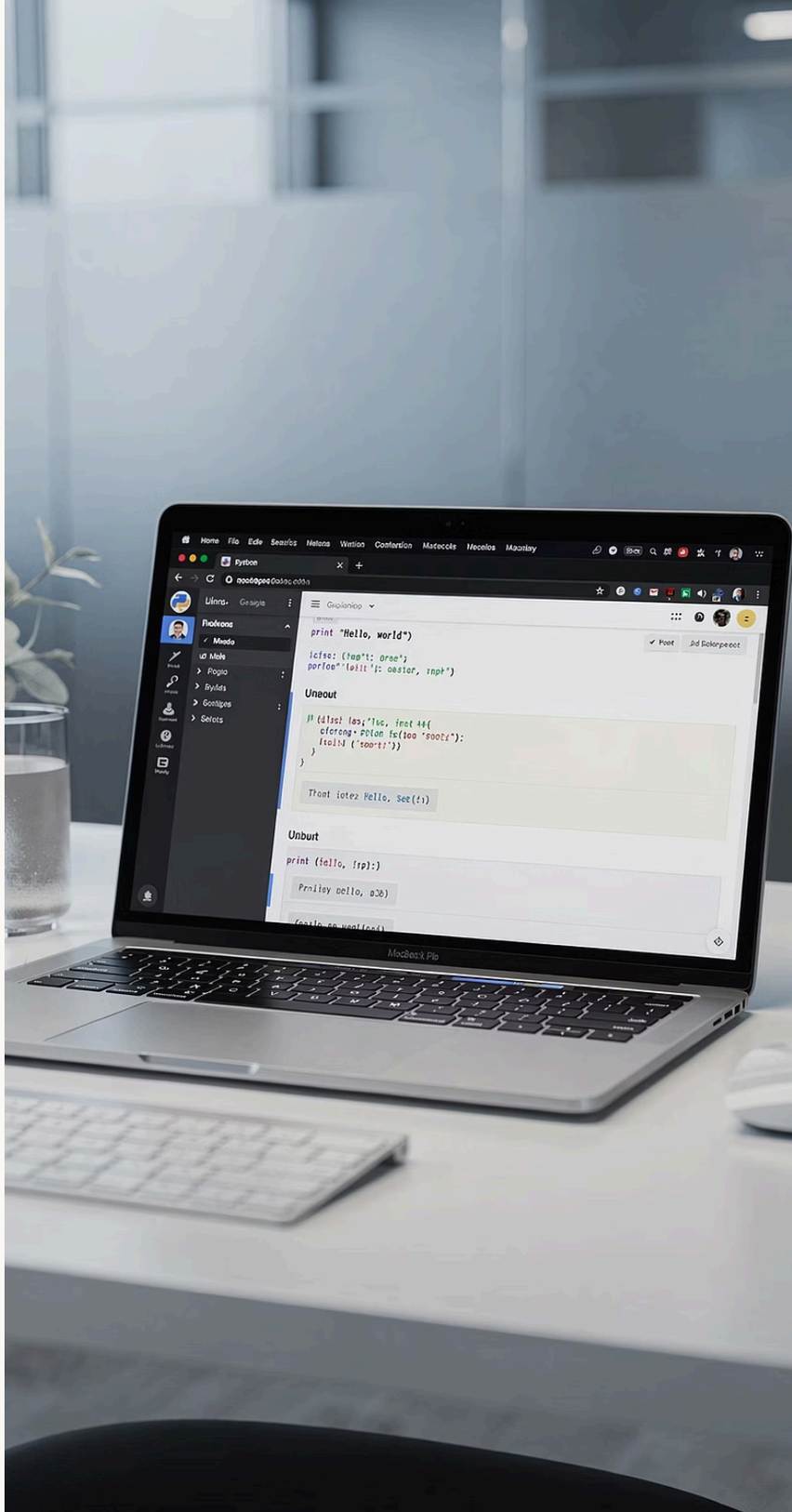
03

✨ WHY START WITH COLAB?

- Zero setup friction (no installation headaches)
- Code in 30 seconds
- Share notebooks instantly
- Free computing power
- Perfect for learning

> 🏢 **ENTERPRISE REALITY:** 65% of data teams use notebooks. Jupyter, Databricks, Colab all similar. Learn once, use everywhere. This skill transfers directly to work.

🎯 TODAY'S GOAL Run your first Python analysis in 10 minutes!



</> PYTHON CONCEPT 1

Variables Storing Data for Reuse

💡 THE BUSINESS PROBLEM

CFO sets daily revenue target: $50,000. You need this number in 20 different calculations. What happens when target changes to $55,000?

📦 WHAT ARE VARIABLES?

Named containers that store data

```
daily_target = 50000
store_name = "Sydney CBD"
is_profitable = True
```

♻️ WHY THIS MATTERS

- **REUSABILITY:** Use same value multiple times
- **MAINTAINABILITY:** Change once, updates everywhere
- **CLARITY:** Names explain what data means

🚫 WITHOUT VARIABLES

Calculate 50000 * 1.1 everywhere

Find and replace 50000 → 55000 (100+ places)

High risk of missing some, breaking calculations

✅ WITH VARIABLES

daily_target = 55000

Every calculation automatically updates!

💼 **BUSINESS IMPACT** If CFO changes target from $50K to $55K: Manual: 30 minutes finding/replacing values. Python: 5 seconds changing one variable

PYTHON CONCEPT 2

Data Types The Right Tool for the Right Job

🔢 WHY DATA TYPES MATTER: Different business data needs different handling

1️⃣ STRING (text)

```
store_name = "Sydney CBD"
manager = "Sarah Chen"
```

2️⃣ INTEGER (whole numbers)

```
transaction_count = 1250
employee_count = 25
```

3️⃣ FLOAT (decimals)

```
revenue = 52847.50
tax_rate = 0.10
```

4️⃣ BOOLEAN (True/False)

```
target_met = True
needs_attention = False
```

⚠️ THE DANGER OF MIXING TYPES

'100' + '200' = '100200' (text)

100 + 200 = 300 (math)

Wrong type = Wrong calculation = Wrong decision!

💼 REAL-WORLD IMPACT

Excel auto-converts types (sometimes wrong)

Python forces you to be explicit (prevents errors)

Correct data types = Accurate business insights

PYTHON CONCEPT 3

Lists Analyzing Multiple Items at Once

📊 THE BUSINESS CHALLENGE

CFO: "Show me top 10 store revenues"

Without lists

10 separate variables

```
store1_revenue = 52000
store2_revenue = 48000
```

...impossible to analyze together!

📋 THE LIST SOLUTION

```
revenues = [52000, 48000, 45000, 42000, 38000]
```

All data in ONE variable!

✨ INSTANT ANALYSIS POWER

Total: sum(revenues)

Average: sum(revenues) / len(revenues)

Highest: max(revenues)

Lowest: min(revenues)

⏱️ **TIME SAVINGS** Manual calculator: 5-10 minutes for 10 stores. Python lists: 0.001 seconds for ANY number!

📈 SCALABILITY

Same code works for:

- 10 stores (today)
- 150 stores (RetailCo)
- 1,500 stores (enterprise)

💼 BUSINESS VALUE: Analyze groups, not just individual items. Process unlimited data with same effort. Foundation for all data analysis

PYTHON CONCEPT 4

Dictionaries Real Business Data Structures

🏪 REAL STORES HAVE MULTIPLE ATTRIBUTES: Name, Location, Revenue, Employees, Manager...

❌ Lists can't connect these together!

```
revenues = [52000, 48000]
names = ["Sydney", "Melbourne"]
```

Which revenue goes with which store? 🤔

✅ 🗂️ DICTIONARY SOLUTION

```
store = {
  "name": "Sydney CBD",
  "revenue": 52847.50,
  "employees": 25,
  "manager": "Sarah Chen"
}
```

All related data in ONE structure!

🔑 KEY-VALUE PAIRS

Access by meaningful names:

store["name"] → "Sydney CBD"

store["revenue"] → 52847.50

💼 CRITICAL INSIGHT

THIS IS HOW DATABASES WORK!

SQL table = Collection of dictionaries

Each row = One dictionary

Each column = One key

🚀 **ENTERPRISE REALITY:** 90% of business data is key-value structured. APIs return dictionaries (JSON). This is THE structure of modern data

Learn dictionaries = Learn database fundamentals

Calculations Automating Business Math

💼 THE CFO'S DAILY QUESTIONS

- What's our total revenue?
- Average per store?
- Who's best/worst performer?
- Are we meeting targets?

🧮 MANUAL PROCESS

- Pull up calculator
- Type each number
- Add/divide manually
- 5 minutes per question
- 20 questions/day = 100 minutes

🐍 PYTHON PROCESS

```
total = sum(revenues)
average = total / len(revenues)
best = max(revenues)
worst = min(revenues)
```

ALL answers in 2 seconds!

98min

Daily Time Saved

100 minutes manual vs 2 seconds Python

426hrs

Annual Hours Saved

= 10 full work weeks!

$27.7K

Cost Savings

426 hours × $65/hour

🎯 **THE REAL POWER:** Same calculations work on 5 stores or 5,000 stores. Change data → Python recalculates instantly. No human errors, ever!

PYTHON CONCEPT 6

If/Else Logic Automated Decision Making

🎯 THE BUSINESS RULE

Categorize 150 stores automatically:

☆ **Excellent: ≥110% target**
→ Bonus eligible

👍 **Good: 100-109%**
→ Maintain performance

! **Attention: 90-99%**
→ Manager coaching

🔕 **Critical: <90%**
→ Immediate intervention

🤖 PYTHON AUTOMATION

```
if revenue >= target * 1.10:
    status = "EXCELLENT"
elif revenue >= target:
    status = "GOOD"
elif revenue >= target * 0.90:
    status = "ATTENTION"
else:
    status = "CRITICAL"
```

✅ WHY THIS MATTERS

- **CONSISTENCY:** Same criteria for ALL stores
- **SPEED:** 150 stores in 2 seconds vs 5 hours
- **ACCURACY:** No human judgment errors
- **SCALABILITY:** Works on any number of stores

⏱️ TIME COMPARISON

Manual: Regional manager reviews each store

150 stores × 2 minutes = 5 hours

Python: Evaluates all stores

150 stores = 2 seconds

💼 BUSINESS IMPACT: Faster problem identification = Higher revenue. Proactive intervention vs reactive cleanup. 15% reduction in revenue loss estimated

PYTHON CONCEPT 7

Loops The Automation Superpower

> ## 🔄 THE SCALING CHALLENGE
>
> RetailCo has 150 stores. Each needs performance evaluation. Manual: 150 × 2 min = 5 hours

🌟 THE LOOP SOLUTION

```
for store in all_stores:
    evaluate_performance(store)
```

Write ONCE, runs on ALL stores!

100%

1 store: 0.01 seconds

100%

150 stores: 2 seconds

100%

1,500 stores: 2 seconds

100%

15,000 stores: 2 seconds!

⚡ THE MAGIC

- Write code for 1 store
- Loop applies it to 150 stores
- Same code handles 1,500 stores
- Same code handles 15,000 stores

💼 CAREER DEFINING MOMENT

THIS is the moment you go from:

❌ Excel data entry clerk

✅ Python automation professional

THIS is what separates:

❌ $75K manual analyst

✅ **$100K+ automation expert**

🚀 THE POWER: Loops = Scale without limits. Write once, run infinitely. The ultimate productivity multiplier!

Putting It All Together

RetailCo Complete Analysis

1️⃣ LOAD DATA

Store 150 store records in dictionaries

2️⃣ CALCULATE METRICS

Total revenue, averages, variances

3️⃣ EVALUATE PERFORMANCE

Apply business rules with if/else

4️⃣ PROCESS ALL STORES

Loop through 150 stores automatically

5️⃣ GENERATE REPORT

Executive summary with action items

⏱️ Manual Process:

- Day 1: Collect data from 150 stores
- Day 2: Calculate metrics, categorize
- Day 3: Create PowerPoint report
- **Total: 3 days (24 work hours)**

⏱️ Python Process:

- 5 min: Load data file
- 2 sec: Run analysis
- 10 min: Format executive summary
- **Total: 15 minutes**

$1.5K

Savings Per Report

$1,560 manual vs $16.25 Python

$30.9K

Monthly Impact

20 reports/month

$370K

Annual Impact

Total business value

🎊 THE CFO'S REACTION

"This is exactly what I needed, instantly!"

"Can you do this for all departments?"

"How soon can you train the team?"

You just became indispensable! 🚀